

Define knowledge

Knowledge in AI **is the** information stored in an AI system **that helps it** think, understand, and make decisions.

Example:

- Fact: **Ravi is a student.**
- Rule: **Students study.**
- Conclusion: **Ravi studies.**

So, **Knowledge = Facts + Rules + Relationships used for reasoning.**

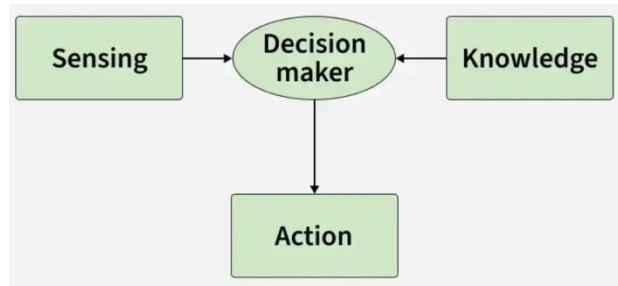
Knowledge Representation in AI

Knowledge Representation (KR) in AI focuses on how machines store and organize real-world information so they can reason, learn, and make intelligent decisions like humans.

- Represents knowledge in a structured form that computers can process.
- Helps AI systems perform reasoning and problem-solving tasks.
- Enables intelligent applications like medical diagnosis and language understanding.
- Allows machines to use stored knowledge and past experiences effectively.

Relation between Knowledge and Intelligence

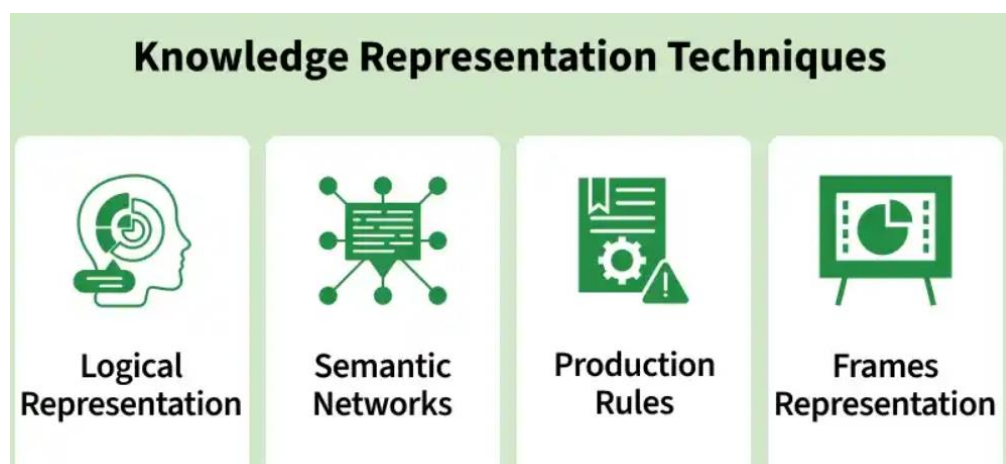
Knowledge is essential for intelligence in both humans and AI systems. AI agents use knowledge and past experience to understand situations, make decisions, and respond intelligently.



The diagram shows that intelligent behavior is possible only when a decision-maker uses knowledge from the environment. This relation can be understood as follows:

- Knowledge about the real world is essential for building intelligent AI systems
- AI agents use knowledge to understand inputs and take appropriate actions.
- An agent can respond correctly only if it has knowledge or prior experience.
- Without knowledge, an AI system cannot display intelligent behaviour.

Techniques of knowledge representation



1. Logical Representation

Logical representation uses formal rules and logic to represent knowledge in AI and helps systems draw conclusions based on given conditions. It includes the following ideas:

- **Syntax:** Rules that decide **rules for writing logical statements** correctly
- **Example:**
Human(Ravi) is a correctly formed predicate statement.
- **Semantics:** Rules that give meaning to logical sentences.
- Human(Ravi) means **Ravi is a human.**

Types of Logic

- **Propositional Logic:** Represents facts as propositions that are either **True or False**.
- **Predicate Logic:** Represents **objects, properties, and relationships** using predicates and variables.

Example: Student Eligibility

Suppose an AI system has the following information:

Fact 1: Ravi is a student.

Fact 2: Ravi has completed all assignments.

Rule: If a student has completed all assignments, then the student is **eligible for the exam.**

Step 1: Represent the facts

Student(Ravi)

CompletedAssignments(Ravi)

Step 2: Represent the rule

Student(x) AND CompletedAssignments(x)

→ EligibleForExam(x)

Here, x represents **any student**.

Step 3: AI performs reasoning

The AI knows:

Student(Ravi) = TRUE

CompletedAssignments(Ravi) = TRUE

The rule says:

If Student AND Completed Assignments

↓

Eligible For Exam

Therefore, the AI concludes:

Eligible For Exam(Ravi)

In Simple Words

Given facts:

Ravi is a student + Ravi completed all assignments

Rule:

If a student completes all assignments → eligible for exam

Conclusion:

Ravi is eligible for the exam.

Advantages

- Helps in logical reasoning and decision making.
- Forms the basis of many programming languages.

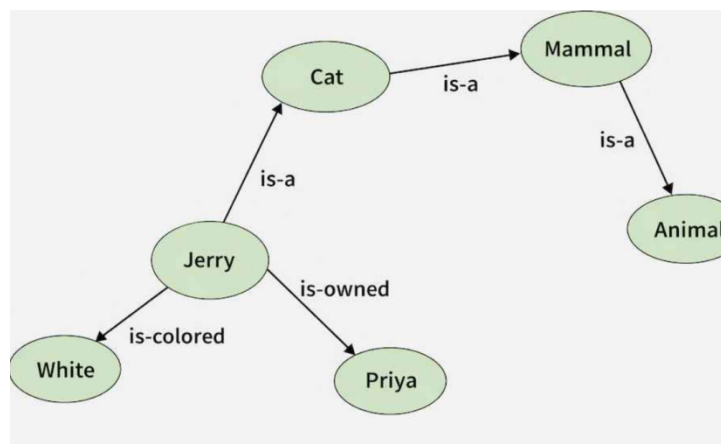
Disadvantages

- Can be difficult to use and understand.
- Reasoning using logic can sometimes be slow or inefficient.

2. Semantic Network Representation

A semantic network represents knowledge using a graph structure with nodes (concepts or objects) and arcs (relationships). It helps show connections between different concepts.

- **IS-A relation:** Shows inheritance (e.g. Cat is a Mammal).
- **Kind-of relation:** Shows category relationships.



Advantages

- Natural and easy way to represent knowledge.

- Simple to understand and visualize.
- Can be easily expanded.

Disadvantages

- Searching large networks can take more time.
- Difficult to create very large networks like human memory.
- Lacks clear standards for link names and logical expressions.

3. Frame Representation

A frame is a data structure used to represent knowledge about an object or situation using slots (attributes) and values. It organizes related information together and is widely used in [NLP](#) and [computer vision](#).

Example: A frame for a Book may include slots such as title, author, year and pages.

Student Frame Representation

Frame	Slot (Property)	Value
STUDENT	Name	Ravi
STUDENT	Age	20
STUDENT	Department	CSE
STUDENT	Year	3
STUDENT	CGPA	8.5

Advantages

- Groups related information together, making programming easier.
- Flexible and easy to extend by adding new attributes.
- Easy to understand and visualize.

Disadvantages

- Inference and reasoning are not always efficient.
- Sometimes too generalized for complex reasoning.

4. Production Rules

Production rules represent knowledge in the form of IF–THEN rules. When a condition is satisfied, the system performs the corresponding action. A production rule system has three main parts:

- Set of rules
- Working memory (current problem state)
- Recognize act cycle (process of checking conditions and applying rules)

Example: IF (bus arrives) THEN (get into the bus).

Advantages

- Easy to understand because rules are written in simple language.
- Highly modular; rules can be added, removed, or modified easily.

Disadvantages

- Systems usually do not learn from past results.
- Many active rules can make the system slower or inefficient.

Applications

- Used in expert systems to provide advice and support decision-making in specialized fields such as medical diagnosis.
- Helps NLP systems understand and generate human language in applications like chatbots and translation tools.
- Enables robots to understand their environment and perform tasks such as navigation and obstacle avoidance.
- Supports the semantic web by helping machines organize and understand relationships between web information.
- Assists cognitive computing systems in analyzing large datasets to generate insights in areas like healthcare research.

Challenges

- Complexity makes it difficult to represent complete domain knowledge accurately.
- Ambiguity occurs when words have multiple meanings that confuse AI systems.
- Scalability becomes challenging when handling very large amounts of data.
- Knowledge Acquisition involves difficulty in collecting and storing expert knowledge.

Discuss PEAS and types of environment with example.

An intelligent agent is a system that perceives its environment through sensors and takes actions through actuators to achieve a goal.

Simple Example: Self-Driving Car

PEAS

PEAS is used to describe **what an AI agent has to achieve, where it works, how it acts, and how it gets information.**

PEAS = Performance Measure + Environment + Actuators + Sensors

P – Performance Measure

It tells us **how we decide whether the agent is successful or not.**

Example: Self-Driving Car

We want the car to:

- Reach the destination safely.
- Avoid accidents.
- Follow traffic rules.
- Use less fuel.
- Reach on time.

These are the **performance measures.**

Question to remember:

"How do we know the agent is doing a good job?"

E – Environment

The **environment** is everything around the agent with which it interacts.

Self-driving car environment:

- Roads
- Other vehicles
- Pedestrians
- Traffic signals
- Buildings
- Weather
- Road signs

Question to remember:
"Where does the agent work?"

A – Actuators

Actuators are the mechanisms through which an agent **performs actions**.

Self-driving car actuators:

- Steering → changes direction
- Accelerator → increases speed
- Brake → decreases/stops speed
- Horn → alerts others
- Indicators → signals turning

Question to remember:
"How does the agent act?"

S – Sensors

Sensors help the agent **collect information from the environment**.

Self-driving car sensors:

- Camera → detects vehicles, people and signs
- GPS → identifies location
- Radar → detects objects and distance
- LiDAR → measures surroundings
- Speed sensor → measures speed

Self-Driving Car

PEAS	Meaning	Example
P – Performance Measure	How success is measured	Safety, reaching destination, following rules
E – Environment	Where agent operates	Roads, traffic, pedestrians, signals
A – Actuators	How agent performs actions	Steering, brakes, accelerator, horn
S – Sensors	How agent receives information	Camera, GPS, radar, LiDAR

Solving the 8-Puzzle Using A* Algorithm

1. Start State

```
+---+---+---+
| 7 | 2 | 4 |
+---+---+---+
| 5 |   | 6 |
+---+---+---+
| 8 | 3 | 1 |
+---+---+---+
```

2. Goal State

```
1 2 3
4 5 6
7 8 _
```

The objective is to arrange the eight numbered tiles in order, with the empty slot in the bottom-right corner.

3. Manhattan Distance

A* uses Manhattan Distance as the heuristic. For every numbered tile, count how many horizontal and vertical grid steps it is away from its correct goal position.

$h(n)$ = Sum of Manhattan distances of all tiles

4. A* Evaluation Function

$f(n) = g(n) + h(n)$

$g(n)$ = cost (number of moves) from the start state to the current state.

$h(n)$ = estimated cost from the current state to the goal using Manhattan Distance.

$f(n)$ = total estimated cost.

5. Generate Possible Moves

```
    2
    ↓
5 → ← 6
    ↑
    3
```

Since the blank is in the center, tiles 2, 5, 6, and 3 can be moved into the blank position. For each resulting state, calculate $g(n)$, $h(n)$, and $f(n)$.

6. A* Search Procedure

Step	Action
1	Start with the initial puzzle configuration.
2	Generate all valid moves of the blank space.
3	Calculate Manhattan Distance $h(n)$ for each new state.
4	Calculate $f(n) = g(n) + h(n)$.
5	Select the state with the lowest $f(n)$.
6	Repeat the process until the goal state is reached.

7. Given Milestones

After Move 8:

2	4	6
7	5	1
—	8	3

After Move 17:

—	6	1
2	5	3
7	8	—

After Move 26 – Goal:

1	2	3
4	5	6
7	8	—

8. Final Answer

The given 8-puzzle can be solved using A* Search with Manhattan Distance. A* repeatedly evaluates the possible states using $f(n) = g(n) + h(n)$ and selects the most promising state. According to the supplied problem statement, the goal state is reached in **26 moves**.

Flow: START → Calculate $g(n)$ and $h(n)$ → Calculate $f(n)$ → Select lowest $f(n)$ → Repeat → GOAL STATE (26 moves)